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GALNT4 promotes the endothelial cell inflammatory response via the NF-κB signaling pathway

Li Wei Guo

071012@xxmu.edu.cn

Xinxiang Medical University

Lulu Zhou

Xinxiang Medical University

Pengcheng Wei

Xinxiang Medical University

Shijie Li

Xinxiang Medical University

Duan Li

Xinxiang Medical University

Research Article

Keywords: GALNT4, Atherosclerosis, Cell adhesion molecules, NF-κB pathway

Posted Date: March 7th, 2024

DOI: https://doi.org/10.21203/rs.3.rs-3998175/v1

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Additional Declarations: No competing interests reported.

Abstract

Background

Atherosclerosis (AS) is a chronic inflammatory disease caused by dysfunction of vascular endothelial cells (ECs). Polypeptide N-acetylgalactosaminyl -transferase 4 (GALNT4) modifies target proteins via O-*N*-acetylgalactosamine (O-GalNAc) glycosylation, which is known to play a crucial role in regulating the inflammatory response in AS, but its exact function in ECs is yet to be determined.

Objective

This study aims to investigate the effect of GALNT4 on endothelial cell inflammation and AS.

Methods and results

We found GALNT4 expression increased in ECs exposed to pro-inflammatory stimuli. GALNT4 over-expression led to upregulation of pro-inflammatory molecules such as ICAM-1, VCAM-1, and MCP-1, which promoted the adhesion of leukocytes to ECs and trans-endothelial migration. Conversely, knockdown of GALNT4 reduced the expression of pro-inflammatory molecules induced by TNF-α. The study also observed that over-expression of GALNT4 increased the binding of NF-κB to the promoter of ICAM-1, VCAM-1, and MCP-1, while GALNT4 knockdown had the opposite effect. Additionally, GALNT4 degraded IκBα and facilitated the translocation of the NF-κB p65 subunit, thereby activating the NF-κB pathway. Finally, GALNT4-mediated endothelial cell inflammation was reduced by the NF-κB inhibitor PDTC and knockdown of the NF-κB p65 subunit, indicating that the NF-κB pathway plays a vital role in regulating GALNT4-mediated expression of adhesion molecules and chemokines.

Conclusion

We provide evidence that GALNT4 promotes the adherence of monocytes to ECs and their trans-endothelial migration via the NF-κB signaling pathway. GALNT4 could be a potential therapeutic target for AS.

Introduction

Cardiovascular disease is a major cause of mortality worldwide and it is primarily caused by atherosclerosis (AS), which is a chronic inflammatory disease of blood vessels. [1, 2]. Inflammation, endothelial dysfunction, and alterations in blood physiology are the main factors contributing to AS[3, 4]. The adhesion and migration of circulating monocytes through vascular walls, based on the expression of adhesion molecules and chemokines, are important characteristics of inflammation reactions[5, 6].

The N-acetylgalactosaminyltransferase 4 (GALNT4) is responsible for initiating the attachment of the first N-acetylgalactosamine (GalNAc) monosaccharide to target proteins during O-linked glycosylation.[7, 8]. It is a member of the GalNAc-T family and is located in the 12q21.33 region, which is linked to coronary artery disease in Chinese Han ancestry [9]. An expression quantitative trait loci (eQTL) study revealed that the lead SNP rs7136259 in the 12q21.33 locus is linked to the expression of GALNT4, which is upregulated in diseased aortic endothelial cells, intimal lesions, and vascular smooth muscle cells of $ApoE^{-/-}$ mice, but downregulated in foam cells[10], suggesting a multi-cellular role for this gene. GALNT4 is believed to play an important role in endothelial-platelet interactions by O-glycosylating the threonine (Thr) residues of the P-selectin glycoprotein ligand (PSGL-1) in monocytes[11]. This process regulates the tethering and rolling of monocytes on ECs and is involved in the inflammatory response in cardiovascular diseases by regulating the modification of PSGL-1 and other inflammation-related proteins[12]. However, the role of GALNT4 in AS has not been studied in ECs.

In this article, we provide evidence that GALNT4 stimulates the interaction of monocytes to the endothelium and up-regulates the expression of the adhesion molecules ICAM-1 and VCAM-1 and the chemokine MCP-1 in ECs. Additionally, the study reveals that the NF-kB pathway is involved in GALNT4-mediated-inflammatory adhesion molecules expression. Therefore, this study is the first to describe the role of GALNT4 in regulating endothelial inflammation and highlights the potent molecular mechanisms involved in this function.

Methods

Cell culture

Primary human umbilical vein endothelial cells (HUVECs) were isolated from human umbilical vein vascular wall by digestion with collagenase (Roche, USA) and cultured in endothelial cell culture medium (ECM) (ScienCell, USA) with 5% fetal bovine serum (FBS), 1% endothelial cell growth supplement (ECGS), 100 U/mL penicillin and 100 mg/mL streptomycin sulfate at 37°C in a humidified incubator with 5% CO_2. Cells from passages 3–6 were used for experiments.

THP-1 and HEK293 cells were obtained from the American Type Culture Collection (ATCC, USA) and cultured in RPMI-1640 medium and DMEM medium supplemented with 10% FBS.

Materials

Recombinant human tumor necrosis factor-alpha (TNF-a) was purchased from PeproTech.Inc (PeproTech, USA). Monoclonal antibodies against GALNT4, IKBα, p65, and Alexa Fluor 488 Anti-rabbit IgG were acquired from Cell Signaling Technology, Inc. (Beverly, MA), Antibodies against β-actin, Goat anti-rabbit and anti-mouse secondary antibodies conjugated to horseradish peroxidase were from Wanleibio (Shenyang, China). A-hICAM-1-APC, a-hVCAM-1-PE, mouse IgG-APC isotype control, and mouse IgG-PE isotype control were purchased from BioLegend.Inc (BioLegend, USA).

Adenovirus construction

A recombinant adenovirus vector expressing the GALNT4 gene (Ad-G4) and an adenovirus containing an empty vector served as a control (Ad-E) was obtained from Hanbio Biotechnology (Shanghai, China). The titers of adenoviruses employed in this study were 2×10^{10} PFU/mL, and the multiplicity of infection (MOI) was 50:1.

siRNA transfection studies

Human GALNT4 siRNA (5'-GGAUUUGUCUCGACCGCUUTT-3'; 5'-AAG CGGUCGAGACAAAUCCTT-3'), human p65 siRNA (5'-GGAUUUGUCUCGA CCGCUUTT-3'; 5'-AAGCGGUCGAGACAAAUCCTT-3') and negative control (NC) siRNAs (TTCTCCGAACGTGTCACGT) were synthesized by GenePharma biotechnology (Shanghai, China). siRNAs were transfected into cells at a final concentration of 100 nM using Lipofectamine RNAiMAX (Invitrogen, USA) following the manufacturer's instructions.

RNA isolation and qRT-PCR

Total RNA was isolated from HUVECs using TRIzol reagent (Invitrogen) according to the manufacturer's protocol. Genomic DNA contamination was eliminated by TURBO DNA-free (Ambion, USA), and cDNA was obtained using the Transcriptor First Strand cDNA Synthesis Kit (Roche, USA). GAPDH was used for normalization. PCR primer sequences are listed in Supplementary **Table** .

Western blot analysis

Total protein was extracted using Cell lysis buffer for Western and IP (Beyotime, China), and the cytoplasmic and nuclear fractions were extracted by the Nuclear and Cytoplasmic Protein Extraction Kit (Beyotime, China) following the manufacturer's instructions. Protein samples were separated by 10% SDS-PAGE and electroblotted to PVDF membranes. The blots were blocked with 5% non-fat dry milk in TBST, and then incubated overnight at 4 ℃ with appropriate primary antibodies. After TBST washing, membranes were incubated with a horseradish peroxidase-conjugated secondary antibody diluted 1:5000 for 1 h at room temperature. The blots were scanned and the intensity of the protein bands was quantified using an image-processing program.

Flow cytometry

Surface analysis of cell membrane proteins was done using direct staining procedures and flow cytometry. The cells were harvested at the end of the experiment using trypsin. They were washed two times in in phosphate-buffered saline (PBS) and re-suspended in 100 uL of ice-cold PBS with 5 uL primary antibody conjugated to APC and PE, Cells were incubated 20 min at 4 ℃ in the dark, then washed in PBS and analyzed by flow cytometry (BD Accuri C6; BD Biosciences). Twenty thousand cells were analyzed for each sample and the expression levels of ICAM-1 and VCAM-1 were expressed as mean fluorescence intensity.

Electrophoretic mobility shift assay (ELISA)

MCP-1 levels were examined using an MCP-1 ELISA kit purchased from R&D Systems (R&D, USA). Samples were analyzed for MCP-1 protein according to the manufacturer's instructions.

Monocyte adherence assay

Human monocyte, THP-1 cells were suspended in serum-free medium (1× 10^6/mL) and incubated with Cell Tracker CM-Dii (Invitrogen, USA) for 20 min at 37 ℃. Labeled THP-1 cells were washed twice with a pre-heated serum-free medium and added (5×10^5/mL) to monolayers of HUVECs. After incubation for 30 min, non-adherent THP-1 cells were removed by washing twice with PBS. The number of binding monocytes was analyzed under a fluorescence microscope and counted with the Image Xpress Microsystem (Molecular Devices, USA).

THP-1 cells were incubated at 37℃ with 50 ug/mL MTT for 2 h and then washed three times with fresh medium. The labeled THP-1 cells were co-incubated with HUVECs for 30 mins, non-adherent cells were removed by washing twice with PBS and 400 uL DMSO was added to each well, then absorbance at 490 nm was measured to represent the number of adherent monocytes.

Monocyte trans-endothelial test

HUVECs were seeded at 1.0×10^5 cells/well onto Transwell filters (5 μm pore, 6.5-mm diameter, Corning, USA) previously coated with matrigel and grown to confluence. THP-1 cells (1.0×10^5 cells/100 μL) were washed and resuspended with serum-free medium and added to the upper chamber. Meanwhile, fresh medium with 10% FBS was added in the lower chamber and left to migrate for 18 h at 37°C, then migrated monocytes in the lower well and bottom filter were counted using Muse Cell Analyzer (Merck Millipore, Germany).

ChIP assay

According to the manufacturer's instructions, Chromatin immunoprecipitation (ChIP) assays were performed using a ChIP assay kit (Upstate, Germany). DNA-bound NF-κB subunit was immunoprecipitated by incubating with an antibody directed against P65 overnight at 4°C, and an IP incubating with normal rabbit IgG was used as a negative control. Next, the crosslinking of protein-DNA complexes was reversed and DNA was purified using spin columns following the manufacturer's instructions. The purified DNA was amplified 40 cycles via PCR to identify the promoter region containing NF-κB binding. The sequence of the primers used to identify the ICAM-1, VCAM-1, and MCP-1 promoters bound to NF-κB are listed in Supplementary **Table 1**.

NF-κB translocation

HUVECs were seeded at 1.0×10^5 cells/well onto glass coverslips in 6-well cell culture plates and grown to confluence. Coverslips were washed twice with PBS, fixed with 4% paraformaldehyde for 15 min at room temperature, and permeabilized with 0.4% Triton X-100 in PBS three times (10 min each). After incubation in blocking buffer (5% BSA in PBS) for 1h, the permeabilized cells were incubated with 1:200 anti-NF-κB-p65 antibody (Cell Signaling Technology, USA) overnight at 4°C. The coverslips were then washed twice

with PBS for 5 min each time and stained with Alexa Fluor 488 Anti-rabbit IgG (CST) diluted in 1:300 for 1 h at room temperature in the dark. Again, the coverslips were washed with PBS three times (10 min each), and cell nuclei were stained with DAPI for 5 min. Finally, coverslips were washed and fluorescent images were observed with a laser-scanning confocal microscope.

Statistical analysis

Statistical analysis was performed using the SPSS17.0 software program. Data were presented as mean ± SEM of at least three independent experiments. Student's t-test was used to compare differences between two conditions, and one-way ANOVA with Bonferroni correction was used for multiple comparisons. All tests were two-sided and significance was assumed at a *P*-value of less than 0.05.

Results

Increased expression of GALNT4 in HUVECs treated by TNF-a in vitro

To investigate the potential role of GALNT4 in endothelial function, we performed the TNF-a-treated HUVECs model *in vitro*. As analyzed by qRT-PCR, results revealed that mRNA expression of GALNT4 was rapid and peaked after 3 h for TNF-a treatment (2.05 ± 0.04 fold) (Fig. 1A), The up-regulation in GALNT4 mRNA coincided with a sustained increase in GALNT4 protein expression following TNF-a treatment (1.2 to 2.3-fold from 6 h to 24 h) (Fig. 1B). Meanwhile, we also found the transcription levels of ICAM-1, VCAM-1and MCP-1 were rapidly and peaked after 3 h for TNF-a treatment (Fig. 1C), the same way as GALNT4. Moreover, we previously tested the expression of GALNT4 in ApoE$^{-/-}$ mouse artery samples, and our results showed that GALNT4 expression increased significantly in the atherosclerosis group compared with the control group (Supplemental **Fig.S1 A and B)**. In addition to TNF-α, other pro-inflammatory stimuli such as LPS (10 ng/mL) and IFN-γ (10 ng/mL) can up-regulate GALNT4 expression in HUVECs (Supplemental **Fig. S1 C and D**).

Taken together, these data indicated that pro-inflammatory stimuli in HUVECs activated GALNT4, and the way TNF-α up-regulated GALNT4 was similar to that it induced pro-inflammatory adhesion molecules, suggesting that GALNT4 may modulate the inflammatory response in ECs.

Over-expression of GALNT4 promotes ICAM-1, VCAM-1, and MCP-1 expression in HUVECs

An important hallmark of EC activation in response to inflammatory stimuli is the fast-peaked expression of pro-inflammatory adhesion molecules. Thus, we looked at the effects of GALNT4 on adhesion molecule expression. The efficiency of adenovirus-mediated over-expression was analyzed and confirmed by qRT-PCR (Fig. 2A) and Western blot (Fig. 2B). Our data suggested that Ad-GALNT4 induced a marked increase of ICAM-1(4.41 ± 1.01 fold), VCAM-1(2.28 ± 0.87 fold), and MCP-1(2.43 ± 0.71 fold) mRNA expression (Fig. 2C-2E), and consistent with these results, Ad-GALNT4 up-regulated the protein secretion

of ICAM-1, VCAM-1 and MCP-1 by 1.95 ± 0.21 fold, 1.76 ± 0.45 fold and 1.53 ± 0.65 fold respectively (Fig. 2F-2H).

Together, these results suggested that enhanced GALNT4 expression promoted ICAM-1, VCAM-1, and MCP-1 expression in HUVECs.

GALNT4 knockdown attenuates TNF-α induced ICAM-1, VCAM-1, and MCP-1 expression in HUVECs

To further confirm the role of GALNT4 in endothelial function, we also applied siRNA technology to knock down the expression of GALNT4. The specific siRNA against GALNT4 results in about 65% knockdown in the expression of GALNT4 at both mRNA and protein levels (Fig. 3A and **B**). Our results demonstrated that knockdown of GALNT4 inhibited the TNF-α-induced expression of ICAM-1, VCAM-1 and MCP-1 by 49.3 ± 0.41% fold, 46.2 ± 0.31% fold and66.0 ± 0.19% fold respectively at the mRNA level (Fig. 3C, E and **D**). And siRNA-GALNT4 down-regulated the protein levels of ICAM-1, VCAM-1, and MCP-1 by 48.9 ± 0.35%, 38.5 ± 2.3%, and 42%± 6.2%, respectively (Fig. 3F,3G and **H**). These results confirmed that endogenous GALNT4 was required to maintain the excessive up-regulation of adhesion molecules in response to the pro-inflammatory stimuli.

GALNT4 promotes adhesion of leukocytes to ECs and trans-endothelial migration

To elucidate the effects exerted by GALNT4-induced cell adhesion molecules, we performed leukocytes-EC adhesion assays. As assayed by DM-Cil labeled monocyte counting, data shown in Fig. 4A and 4B, when we over-expressed GALNT4 in HUVECs, the number of THP-1 adhering to the endothelial cell monolayer was significantly increased by1.57 ± 0.59 fold in basal and 2.01 ± 0.38 fold in TNF-α-induced conditions. The siRNA-mediated knockdown significantly reduced the TNF-α-induced leukocyte adhesion to EC monolayers by 35 ± 1.9%. The same results were found by MTT labeled monocyte OD490nm absorbance value, Ad-GALNT4 increased leukocyte adhesion to ECs by 1.51 ± 0.13 fold in basal and 1.73 ± 0.43 fold in TNF-α-induced conditions, and GALNT4 knockdown decreased the TNF-α-induced leukocyte adhesion to ECs by 32.7 ± 0.91% fold respectively (Fig. 4C).

Meanwhile, we examined the effect of GALNT4 on the leukocyte's trans-endothelial migration ability. THP-1 cells that transmigrated through the matrigel-membrane unit and attached to the bottom filter were estimated by flow cytometry and cell counting (Fig. 4D). The data demonstrate that the trans-endothelial migration of THP-1 cells was greatly promoted by Ad-GALNT4 by 1.52 ± 0.62 fold in basal and 2.06 ± 0.73 fold in TNF-a stimulation. However, GALNT4 knockdown significantly impaired the positive effect of TNF-a by 33.5 ± 1.5%(Fig. 4D).

Taken together, these results show that GALNT4 promotes cell adhesion and trans-endothelial migration simultaneously.

GALNT4 activates NF-κB signaling pathway in HUVECs

To shed light on the signaling pathway induced by GALNT4, we sought to assess whether GALNT4 can activate NF-κB in HUVECs. First, we performed a quantitative analysis of the NF-κB p65 binding activities *in vitro* by ChIP analysis. Immunoprecipitated chromosomal DNA with p65 antibodies was subjected to PCR, using primers designed to amplify the ICAM-1, VCAM-1, and MCP-1 promoter region harboring NF-κB transcription factor binding sites. As shown in Fig. 5A, *In vivo* binding of NF-κB to the ICAM-1, VCAM-1 and MCP-1 promoter occurred after TNF-αstimulation. The binding activities of NF-κB were enhanced by Ad-GALNT4 while reduced by GALNT4 knockdown.

Next, we determined the effect of GALNT4 on the expression of the main components of the NF-κB pathway, including p65 subunit and total IκBα. We did not observe any significant differences between GALNT4 over-expression and silencing on the total p65 subunit levels (data shown in Supplemental Fig. 2A **and B**); however, we found that HUVECs over-expressing GALNT4 had significantly more IκBα degradation in their cytoplasm (Fig. 5B). After the proteasomal degradation of IκBα and its release from the IκB complex, translocation of NF-κB from the cytoplasm to the nucleus occurs, an essential step for the activation of NF-κB target genes. We further explored the effect of GALNT4 on this process by measuring the p65 protein levels in the cytoplasmic and nuclear fractions. As shown in Fig. 5C, GALNT4 over-expressing cells showed increased p65 levels in nuclear fractions, while reduced p65 levels were detected in nuclear fractions from GALNT4-silenced cells (Fig. 5D). Furthermore, the decreased relocation of p65 into the nuclei in GALNT4-silenced cells was confirmed by confocal microscopy (Fig. 5E).

Taken together, these results demonstrated that GALNT4 positively activates NF-κB pathway activity in HUVECs.

NF-κB signaling pathways involved in GALNT4-mediated ICAM-1, VCAM-1, and MCP-1 expression

To determine whether NF-κB participates in the GALNT4-mediated adhesion molecule and chemokine expression, the involvement of the NF-κB signaling pathway in GALNT4-induced ICAM-1, VCAM-1, and MCP-1 expression was investigated. Decreased levels of ICAM-1, VCAM-1, and MCP-1 expression were initially detected in HUVECs following pre-treatment with the NF-κB inhibitor PDTC both at mRNA level and protein level. (Fig. 6A–6D). The effect of siRNA against p65 and the efficiency of the p65 siRNA was confirmed by mRNA and protein analyzed through RT-PCR and western blot (data shown in supplement **Fig.S2 C and D**). HUVECs were transfected with p65 siRNA overnight before being infected with Ad-GALNT4. As shown in Fig. 6E–6G, transfection with p65 siRNA significantly reduced the ICAM-1, VCAM-1, and MCP-1 up-regulation induced by GALNT4 over-expression. In addition, we measured monocyte adhesion to HUVECs after treatment with PDTC or transfection with p65 siRNA. Both the pretreatment and transfection significantly inhibited the amount of monocyte adhesion and monocyte trans-endothelial migration (Fig. 6H and 6K).

Together, these results suggest that GALNT4 regulates ICAM-1, VCAM-1, and MCP-1 expression through the NF-κB signaling pathway in HUVECs.

Discussion

In the present study, we demonstrate for the first time the function of GALNT4 in HUVECs. We found that the expression of GALNT4 increases in response to pro-inflammatory stimuli and it induces the expression of adhesion molecules such as ICAM-1 and VCAM-1, as well as chemokines like MCP-1. This mechanism regulates monocyte adhesion to HUVECs and trans-endothelial migration through the NF-κB signaling pathway.

Glycosylation is a common post-translational modification in proteins because about half of the known proteins in eukaryote cells are glycosylated[13, 14]. There are two main types of protein glycosylation: N-linked and O-linked. O-linked glycosylations are subdivided into mucin-type and O-GlcNAcylation[15–17]. Mucin-type O-glycosylation is initiated by adding GalNAc to Ser or Thr residues of proteins in the Golgi[18]. This process in mammals is initiated and regulated by a large family of 20 GalNAc-Ts (EC 2.4.1.41) [8], GALNT4 is the fourth human GalNAc-T and recent studies have shown that it plays a role in regulating the cardiovascular system[11, 19, 20]. However, its biological functions are not yet fully understood. Previous research has shown that GALNT4 expression was up-regulated in diseased MAECs, intimal lesions, and SMCs of $ApoE^{-/-}$ mice, but down-regulated in foam cells, indicating a multi-cellular role for this gene[10]. In the present study, GALNT4 expression rapidly increased and peaked after just 3 h of TNF-α treatment in HUVECs *in vitro*, suggesting that GALNT4 is critical to endothelial cell function and its expression is promoted by the initiation of AS.

GALNT4 plays an important role in endothelial-platelet interactions by O-glycosylating the Thr residues of the PSGL-1, which interacts with P-selectin expressed on the surface of endothelial cells. This interaction regulates the tethering and rolling of monocytes on these cells[11, 12]. However, the role of GALNT4 in HUVECs has not been studied before. In this study, we used gain- and loss-of-function approaches to investigate the role of GALNT4 in HUVECs. We found that over-expression of GALNT4 in HUVECs up-regulated the transcription and protein secretion of several cell adhesion molecules, such as ICAM-1, VCAM-1, and MCP-1. Conversely, GALNT4 knockdown dramatically suppressed the TNF-α induced ICAM-1, VCAM-1, and MCP-1 expression. Endothelial cells secrete cell adhesion molecules that activate inflammation in the vascular endothelium, facilitating monocyte adhesion and transmigration[21]. Our study found that GALNT4 dynamically promotes monocyte adhesion to HUVECs and monocyte trans-endothelial migration. Further studies are needed to verify the relevance of these findings in GALNT4 gene knockout models, especially in cell-type-specific knockout models.

The activation of a central transcription factor called NF-κB is induced by TNF-α, which is a crucial regulator of gene expression[22]. This activation usually occurs through signal-induced phosphorylation and proteasome-mediated degradation of IκB. This degradation releases bound NF-κB, allowing it to move to the nucleus where it activates gene transcription[23]. In the present study, we found that over-

expressing GALNT4 resulted in a significant increase in the levels of IκBα degradation in HUVECs, as well as translocation of the p65 NF-κB subunit to the nucleus. These responses were suppressed in GALNT4-knockdown cells. As NF-κB family members are preformed proteins, their initial activation and activity are often regulated by posttranslational modifications rather than by induction of their synthesis. Previous work has shown that NF-κB activity is regulated by several posttranslational modifications[24], including phosphorylation[25], methylation[26], and acetylation[27]. GalNAc-Ts control the initiation of GalNAc-type O-glycosylation, a process that involves GalNAc to Ser and Thr residues, which may occupy the same or adjacent Ser or Thr residues in proteins in proteins, just like phosphorylation modifications[28, 29]. Several lines of evidence suggest a reciprocal relationship between O-GlcNAc and O-phosphate [30], Whether GalNAc-type O-glycosylation functions in the same way needs further investigation.

The NF-κB pathways play an important role in ICAM-1, VCAM-1, and MCP-1 expression in HUVECs[31]. In this study, we used a specific NF-κB inhibitor, PDTC, to examine the role of NF-κB in GALNT4-mediated cell adhesion molecule expression. PDTC inhibited NF-κB activity at a very low dosage (20 μM) but did not affect cell adhesion molecule expression. We found that PDTC significantly inhibited the enhancement of GALNT4-mediated cell adhesion molecule expression. Moreover, we confirmed the role of NF-κB in this process by using p65 siRNA. There are several NF-kB binding sites within the promoter regions of the ICAM-1, VCAM-1, and MCP-1 genes. As shown by a ChIP assay, we found that over-expression of GALNT4 increased the binding of NF-κB within the ICAM-1, VCAM-1, and MCP-1 gene promoters, while knockdown of GALNT4 led to opposite results. Furthermore, PDTC and p65 siRNA also reduced GALNT4-induced monocyte adhesion to HUVECs and monocyte trans-endothelial migration.

In summary, the results of this study demonstrate an essential function of GALNT4 in HUVECs by positively regulating the expression of cell adhesion molecules. GALNT4 promotes NF-κB activity and functionally enhances the NF-κB dependent cell adhesion molecule promoter activity. These results provide new insights into the molecular mechanisms underlying the development of AS.

Declarations

Funding

This work is supported by the National Natural Science Foundation of China (No.81900391) and the Open Program of Henan Key Laboratory of Biological Psychiatry (Program No. ZDSYS2020005).

Author contributions

Li-Wei Guo designed the study. Lu-lu Zhou, and Peng-cheng Wei conducted the experiment, and Shi-jie Li analyzed the data and wrote the manuscript. Duan Li conceived and helped write the manuscript. All authors commented on previous versions of the manuscript, and all authors read and approved the final manuscript.

Conflict of interest

All authors declare that there is no conflict of interest.

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Figures

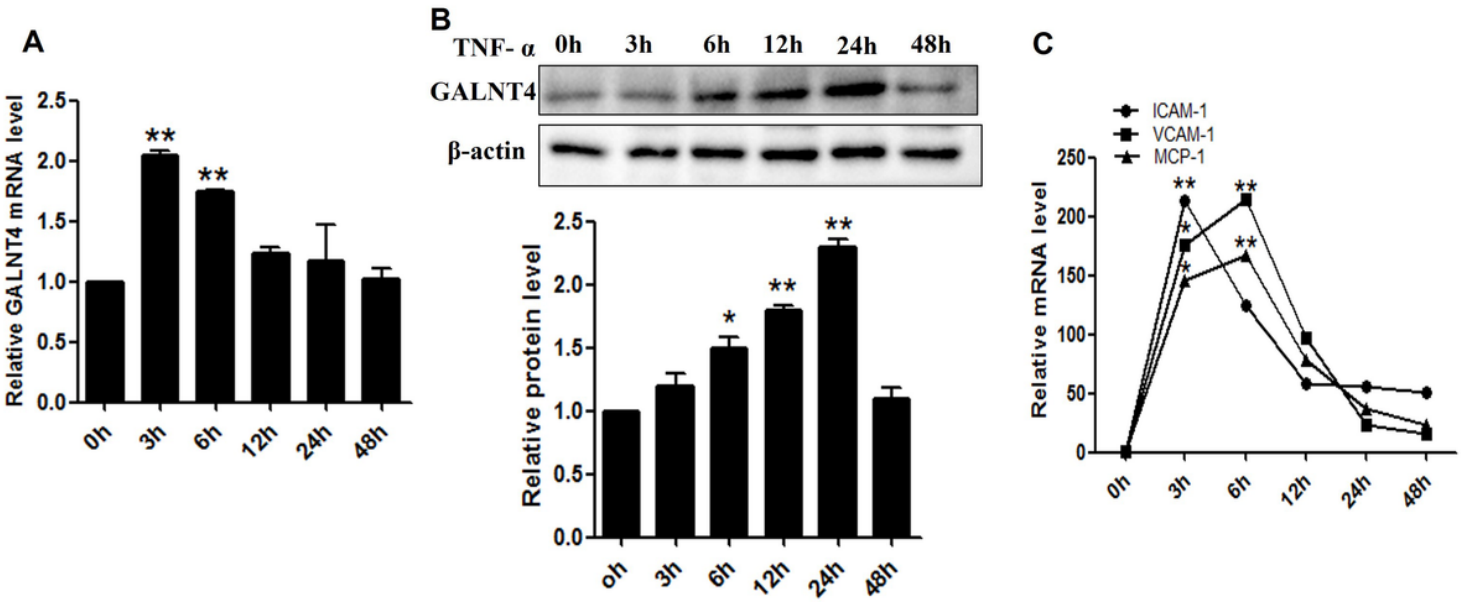


Figure 1

Increased expression of GALNT4 in HUVECs treated by TNF-a *in vitro.*

A-B, HUVECs were treated with TNF-α (10 ng/mL) for the indicated time points. The mRNA expression of GALNT4 was determined by qRT-PCR (**A**), and protein levels by Western blot (**B**). **C**, HUVECs were treated with TNF-α (10 ng/mL) for the indicated time points. the mRNA level of ICAM-1, VCAM-1and MCP-1was determined by qRT-PCR. Data shown here are from three independent experiments and presented as mean ± SEM, **P*<0.05, ***P*<0.01.

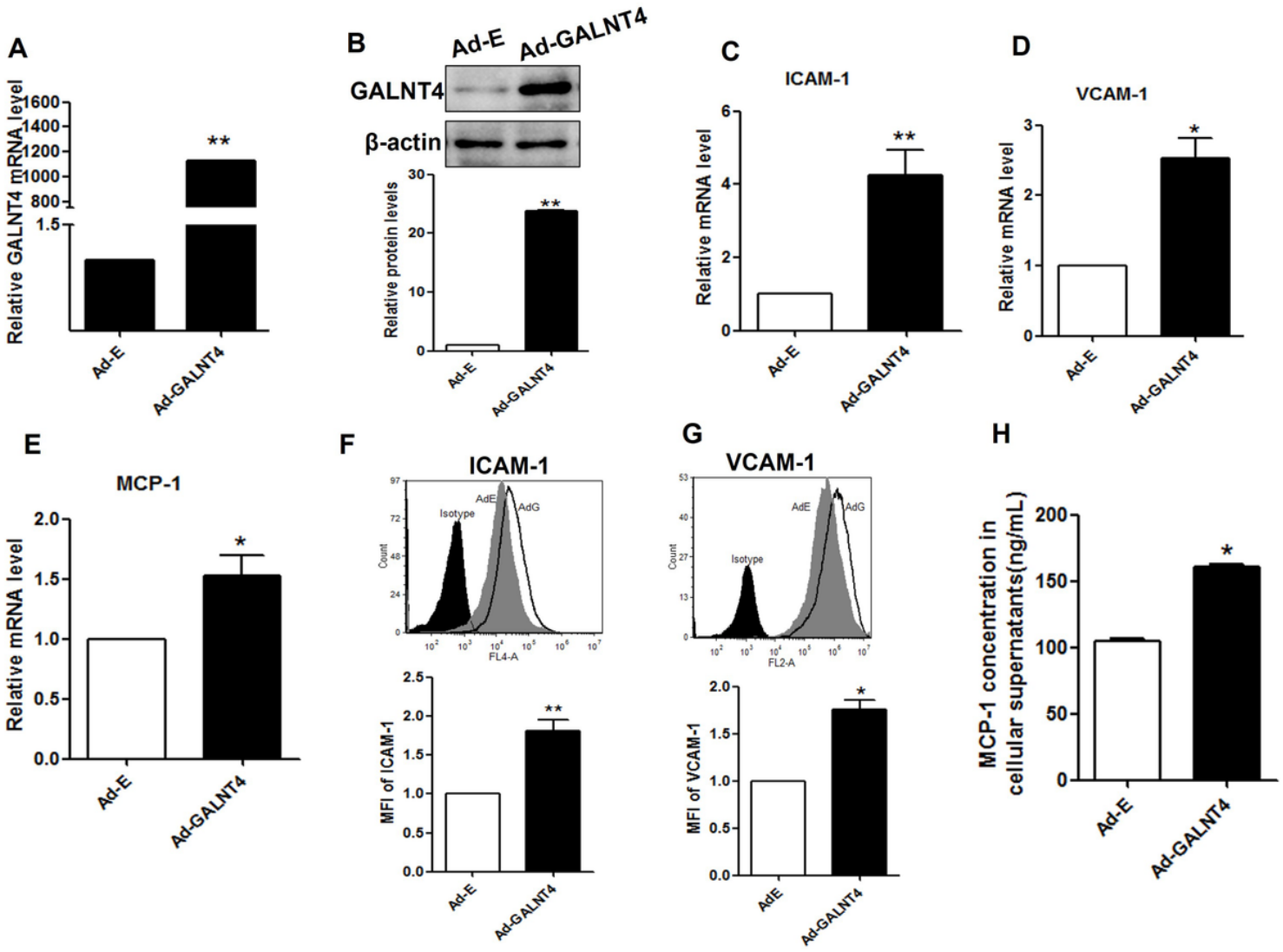


Figure 2

Over-expression of GALNT4 promotes ICAM-1, VCAM-1 and MCP-1 expression.

A-B, HUVECs were infected with Ad-GALNT4 or empty vector (50moi), 48h after transduction, total RNA and protein were extracted for the detection of GALNT4 by qRT-PCR(**A**) western blot analysis (**B**). **C-E**, mRNA expression of ICAM-1, VCAM-1, and MCP-1 was analyzed by qRT-PCR. **F-G**, The APC-conjugated ICAM-1 and PE-conjugated VCAM-1 were used to determine the surface expression of ICAM-1 (**E**) and VCAM-1 (**F**) by flow cytometry; **H**, The concentration of MCP-1 in the supernatants of cultured HUVECs was analyzed by ELISA. The data shown represent the mean ± SEM of three independent experiments, **P*<0.05, ***P*<0.01.

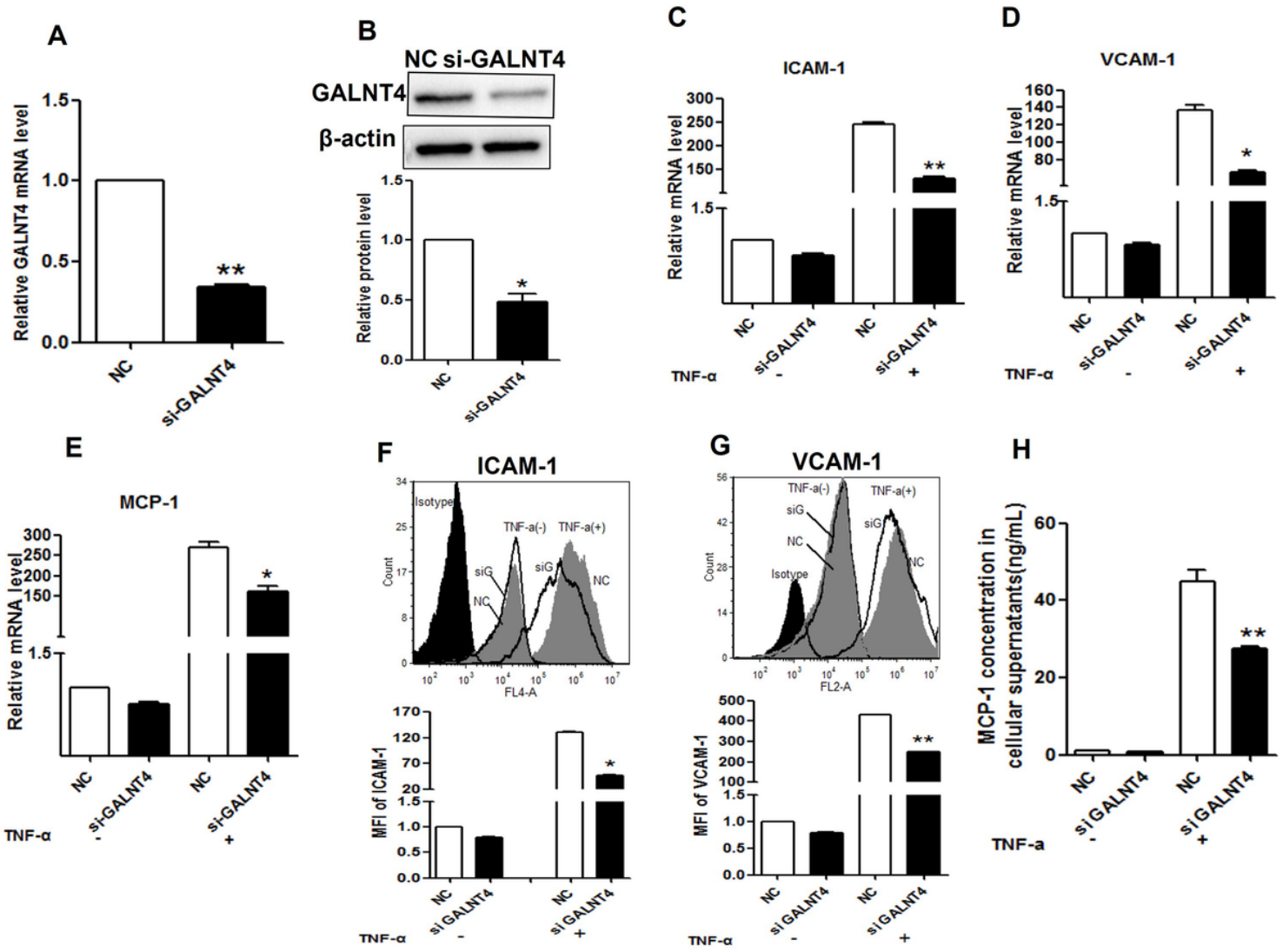


Figure 3

Knockdown of GALNT4 attenuates TNF-α induced ICAM-1, VCAM-1, and MCP-1 expression

A-B, HUVECs were transfected with siRNA-negative control or siRNA-GALNT4 (100 nM) for 48 h, and total RNA and protein were extracted for the detection of GALNT4 by qRT-PCR (**A**) western blot analysis (**B**). **C-E**, HUVECs were transfected with siRNA-negative control or siRNA-GALNT4 (100 nM) for 48 h, and then treated with TNF-α (10 ng/mL) for 2 h or left untreated. The mRNA expression of ICAM-1, VCAM-1 and MCP-1 was analyzed by qRT-PCR. **F-G**, The APC-conjugated ICAM-1 and PE-conjugated VCAM-1 were used to determine surface expression of ICAM-1 (**E**) and VCAM-1 (**F**)by flow cytometry; **H**, The concentration of MCP-1 in the supernatants of cultured HUVECs was analyzed by ELISA. The data shown represent the mean ± SEM of three independent experiments, **P*<0.05, ***P*<0.01.

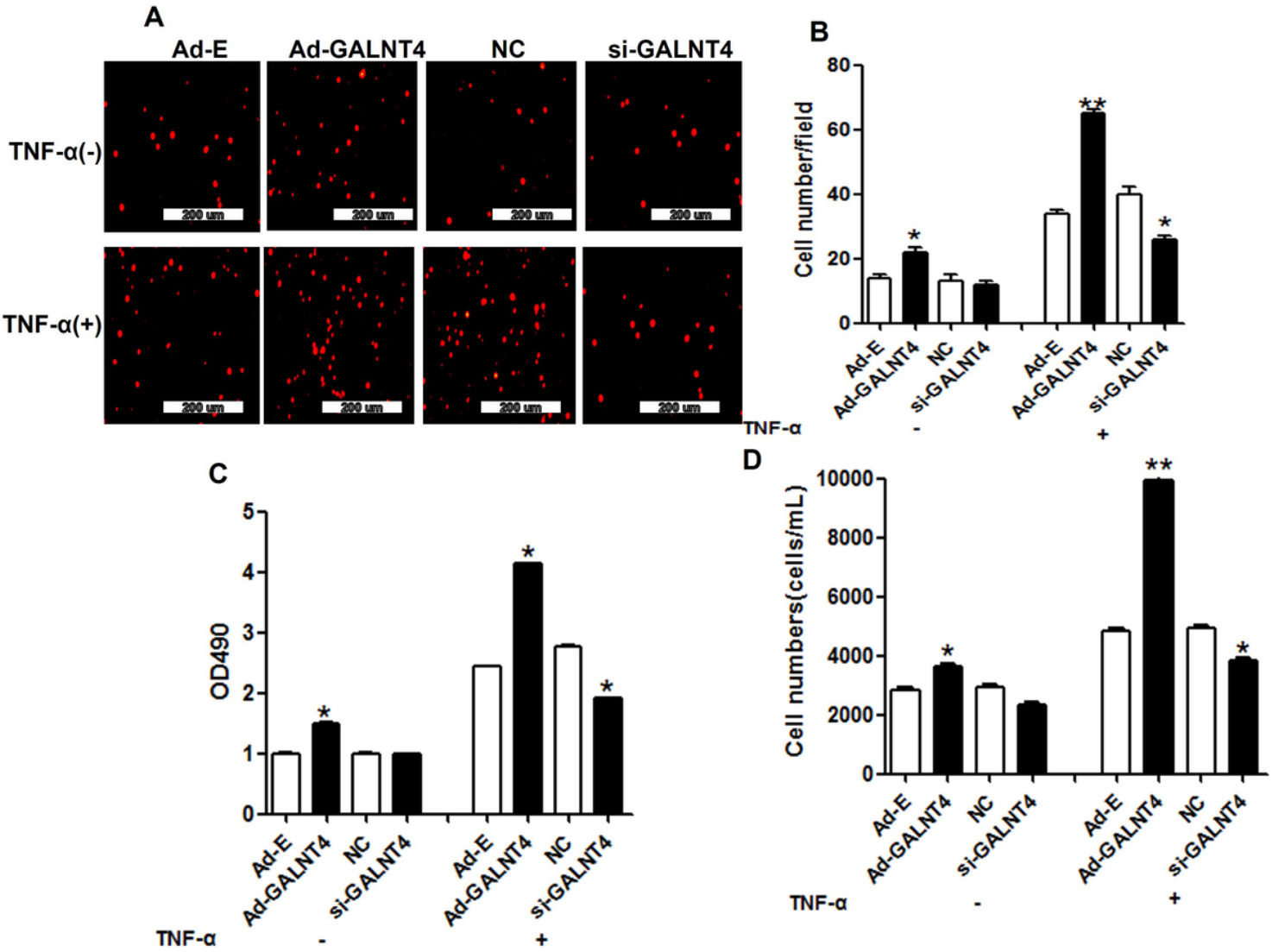


Figure 4

GALNT4 enhances monocyte-EC adhesion and monocyte trans-endothelial migration

A-B, HUVECs were infected with Ad-GALNT4 and empty vector, or transfected with GALNT4 siRNA and control siRNA, with or without 10 ng/mL TNF-α treatment for 2 h. Activated HUVECs were incubated with CM-Dil-labeled THP-1 cells for 30 min, and the adhesion of THP-1 to EC was visualized (**A**) and counted with Image Xpress Microsystem (**B**). **C**, THP-1 cells were pre-stained with MTT for 2 h and co-incubated with HUVECs for 30 min, the average fold change of the absorbance at 490 nm was analyzed. **D**, The confluent monolayer of HUVECs cultured on matrigel-coated transwell filters were activated for the indicated treatment above, THP-1 cells were washed and resuspended with no serum medium and added to the upper well, after 18 h of migration at 37°C, THP-1cells in the lower well and bottom filter were counted using Muse Cell Analyzer. The data shown represent the mean ± SEM of three independent experiments, **P*<0.05, ***P*<0.01.

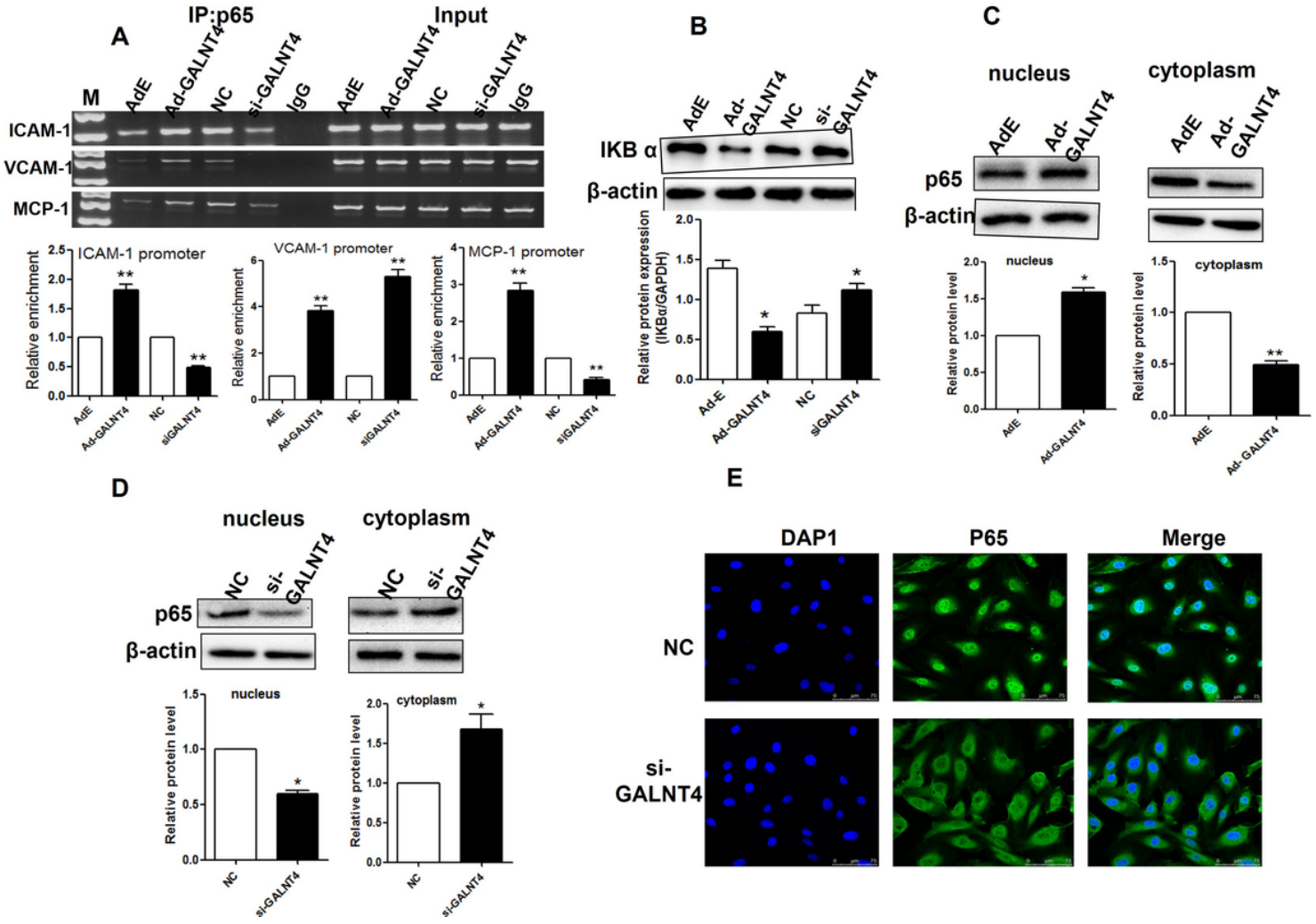


Figure 5

GALNT4 activates NF-κB signaling pathway in HUVEC*s*

A. HUVECs were infected with Ad-GALNT4 and empty vector, or GALNT4 siRNA and control siRNA for 48 h, then stimulated with TNF-α (10 ng/mL) for 2 h, ChIP assays were performed with an antibody against p65 and normal rabbit IgG. The binding of p65 to the ICAM-1, VCAM-1, and MCP-1 promoters was determined with PCR. **B**,HUVECs were infected with Ad-GALNT4 and empty vector, or GALNT4 siRNA and control siRNA for 48 h, total protein was extracted, and IkBa protein level was analyzed by western blot analysis. **C-D**, The indicated proteins were detected in cytoplasmic or nuclear fractions prepared from HUVECs transfected with Ad-GALNT4 and empty vector, or GALNT4 siRNA and control siRNA for 48 h, and then treated with 10 ng/mL TNF-α for 2 h. The fold change of p65 protein expression was normalized to β-actin. **E**, HUVECs were transfected with GALNT4 siRNA and control siRNA for 48 h, and then treated with 10 ng/mL TNF-α for 2 h. Nuclear translocation of the p65 NF-κB subunit was assessed by immunofluorescence with antibodies against p65. The data shown represent the mean ± SEM of three independent experiments, **P*<0.05, ***P*<0.01.

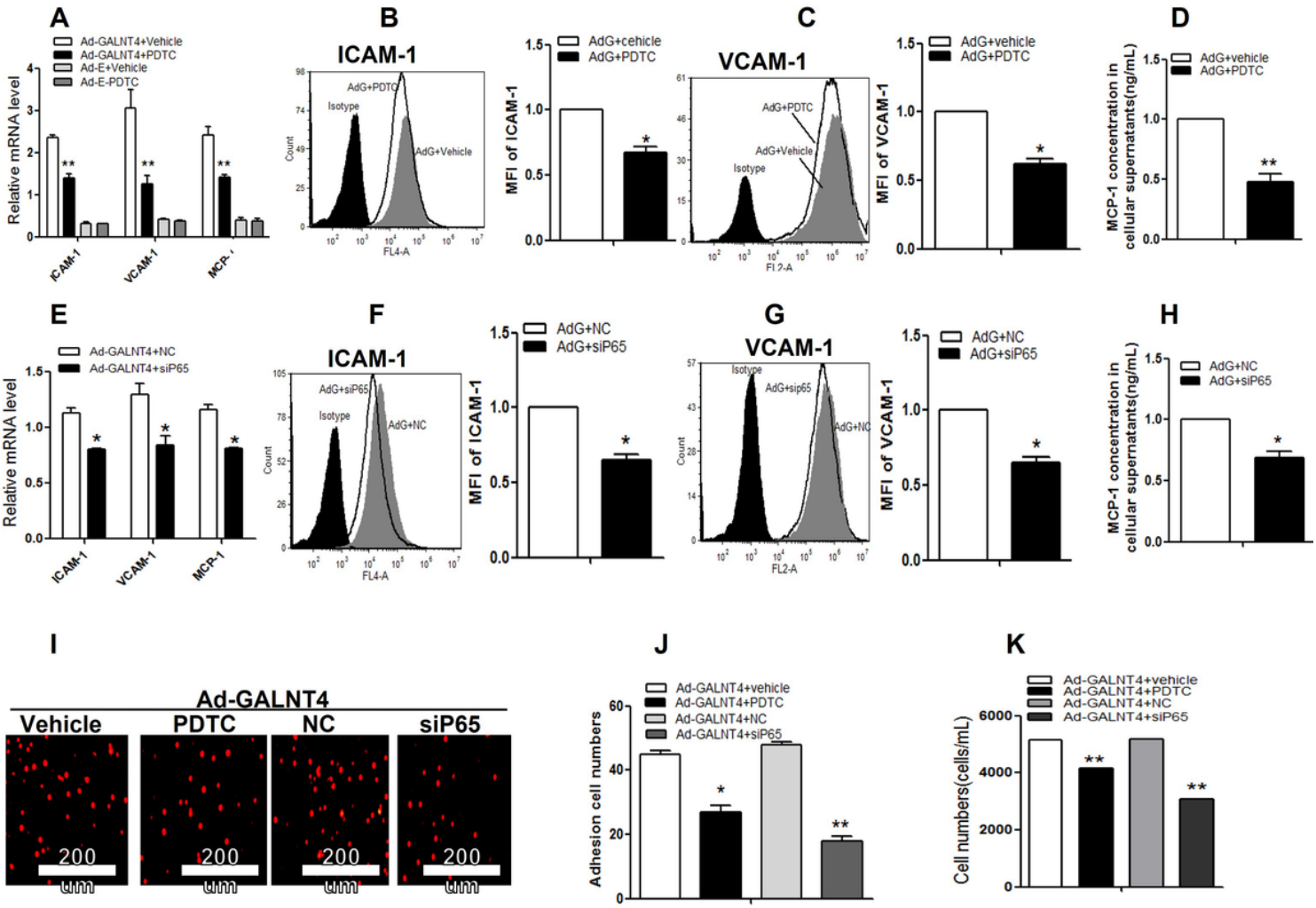


Figure 6

NF-κB pathway is involved in GALNT4-induced ICAM-1, VCAM-1 and MCP-1 production

A, HUVECs were pretreated with NF-κB signal inhibitors PDTC(20 uM) for 2 h, and then infected with Ad-GALNT4 for another 48 h, ICAM-1, VCAM-1 and MCP-1 mRNA expression were analyzed by RT-PCR. **B-C**, ICAM-1, and VCAM-1 physical expression were detected using flow cytometry. **D**, MCP-1 protein expression was analyzed by ELISA. **E**, HUVECs were transfected with p65 specific siRNA or control siRNA for 24 h, then the cells were infected with Ad-GALNT4 for another 24 h, ICAM-1, VCAM-1, and MCP-1 mRNA were analyzed by RT-PCR. **F-G**, the Protein level of ICAM-1 and VCAM-1 were analyzed using flow cytometry. **H**, MCP-1 protein expression was analyzed by ELISA. **I-J**, HUVECs were pretreated with PDTC for 2 h or transfected with p65 siRNA followed by infected with Ad-GALNT4 for 48 h. THP-1 cells labeled with CM-Dil were added to and co-incubated with HUVECs for 30 min, the THP-1 cell's adherence was measured and counted by fluorescence microscopy. **K**, HUVECs were pretreated with PDTC for 2 h or transfected with p65 siRNA followed by infected with Ad-GALNT4 for 48 h.THP-1 cells were washed and resuspended with no serum medium and added to the upper well, after 18 h migration at 37°C, monocytes in the lower well and bottom filter were counted using Muse Cell Analyzer. Results are expressed as the mean±SEM of three independent experiments, ** P<0.01 and * P<0.01.

Supplementary Files

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